

CBE 30235: Introduction to Nuclear Engineering

Problem Set 4: Interactions, Moderation, and Fission

Due: Friday, February 13, 2026, at 11:59 PM

Instructions

- Show all work clearly. State your assumptions.
 - For numerical answers, units are mandatory.
 - Submit as a single PDF on Canvas.
 - **Constants:**
 - Avogadro's Number: $N_A = 6.022 \times 10^{23}$ nuclei/mol
 - 1 barn (b) = 10^{-24} cm²
 - Elementary Charge: $e = 1.602 \times 10^{-19}$ C (therefore 1 eV = 1.602×10^{-19} J)
 - Mass of Neutron ≈ 1.00866 u
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Problem 1: Shielding Design and Mean Free Path

You are designing a biological shield for a neutron beam. The beam consists of thermal neutrons ($E = 0.025$ eV) with an intensity $I_0 = 10^{10}$ neutrons/cm²·s. You decide to use a Boron-10 (¹⁰B) shield because of its massive absorption cross-section ($\sigma_a = 3840$ barns). The density of pure ¹⁰B is $\rho = 2.34$ g/cm³.

- Calculate the **atom density** (N) of the shield in nuclei/cm³.
- Calculate the **macroscopic absorption cross-section** (Σ_a) in cm⁻¹.
- Determine the **mean free path** (λ) of a thermal neutron in this shield.
- How thick must the shield be to reduce the beam intensity to 10^3 neutrons/cm²·s? (Assume simple exponential attenuation).

Problem 2: The Mechanics of Slowing Down

In a thermal reactor, we must slow neutrons from fission energy ($E_0 = 2$ MeV) down to thermal energy ($E_{th} = 0.025$ eV). Compare the performance of three potential moderators: Hydrogen ($A = 1$), Graphite (¹²C), and Iron (⁵⁶Fe).

- Calculate the collision parameter $\alpha = [(A - 1)/(A + 1)]^2$ for all three nuclides.
- If a neutron with $E = 2$ MeV hits a Carbon-12 nucleus, what is the **minimum** energy it could have after a single elastic collision?

- (c) Calculate the average logarithmic energy decrement ξ for all three. (Use the approximation $\xi \approx 2/(A + 2/3)$ for $A > 1$, but use the exact value $\xi = 1$ for Hydrogen).
- (d) Calculate the **average number of collisions** required to thermalize a neutron (from 2 MeV to 0.025 eV) for each material. Based on this, why is Iron a poor moderator despite having a reasonable scattering cross-section?

Problem 3: Reactor Power and Fuel Consumption

This is a mass balance problem. Consider a typical commercial nuclear power plant operating at a **Thermal Power of 3,000 MW_{th}**. (Note: This is heat rate, not electric output). Assume the recoverable energy per fission is exactly **200 MeV**.

- (a) Calculate the **Fission Rate** (fissions per second) required to maintain this power level.
- (b) Calculate the rate of ^{235}U consumption in **grams per day**. (Assume only ^{235}U fissions).
- (c) If the fuel is Uranium Dioxide (UO_2) enriched to 4.5% ^{235}U by weight, how many **kilograms of UO_2 fuel** are consumed (depleted of fissile isotope) per year of continuous operation?

Problem 4: Fission Physics Concepts

Briefly answer the following based on the Liquid Drop Model and the "Satellite Droplet" analogy discussed in class.

- (a) Explain why ^{235}U is **Fissile** (fissions with slow neutrons) while ^{238}U is only **Fissionable** (requires fast neutrons). Use the concepts of **Binding Energy** vs. **Critical Energy** in your answer.
- (b) The mass yield curve for fission is asymmetric, peaking at $A \approx 95$ and $A \approx 140$. Why are these fission fragments highly radioactive? (Explain in terms of the Neutron-to-Proton ratio).
- (c) We discussed that $\sim 80\%$ of fission energy appears as Kinetic Energy of the fission fragments. In terms of heat transfer engineering, **where** exactly is this heat generated in the reactor, and **why** is it deposited there (i.e., what is the range of these fragments)?

Problem 5: Enrichment and the Reproduction Factor η

The feasibility of a chain reaction depends on η (the number of neutrons produced per absorption in the fuel).

$$\eta = \nu \frac{\Sigma_f}{\Sigma_a}$$

Consider a mixture of ^{235}U and ^{238}U . Assume the following microscopic thermal cross-sections and data:

- ^{235}U : $\sigma_f = 582$ b, $\sigma_\gamma = 99$ b, $\nu = 2.42$
- ^{238}U : $\sigma_f = 0$ b, $\sigma_\gamma = 2.7$ b

- (a) Calculate η for pure ^{235}U .

- (b) Calculate η for **Natural Uranium** (which contains 0.72% ^{235}U and 99.28% ^{238}U by atom fraction).
- (c) Calculate η for **Low Enriched Uranium (LEU)** (which contains 4.0% ^{235}U and 96.0% ^{238}U).
- (d) To sustain a chain reaction in water, we typically need $\eta > 1.8$ to overcome leakage and absorption in the moderator. Based on your results, why can't we use natural uranium in a standard Light Water Reactor?